

Course: MS&E 152 summer 2026 Sequence: Week 8 Lecture 1 Date: Monday, August 10, 2026 Topic: Decision Analysis in Practice - Framing the Problem.

Links:

Course website: <https://stanford-msande152.github.io/summer26/> Canvas: <https://canvas.stanford.edu/courses/228284>

Title: The practice: Working with Decision-Makers and Stakeholders

What you will learn

Class schedule

- Lecture:
 - About this class
 - Short break
 - Second lecture
 - Class activity
-

I. Structuring and scoping to frame a decision problem

The initial structuring phase of decision analysis is called "framing," as mentioned in our first lecture on "what is Decision Analysis." Ron Howard wrote that framing is uniquely human. (D. Skinner, Ch.6) It captures what is unique to the problem at hand and cannot be duplicated by a software tool, or a generative AI model. *The goal of the framing phase is to make the problem clear and understood for all participants.* This involves

- Identifying the different parties to engage in the problem,
- Scoping the problem to encompass the issues surrounding the decision context.
- Creating possibly novel alternatives that fully address the decision-maker's values.

The initial phase to define the problem is arguably the most important determinant of success, and the least amenable to formulaic approaches.

Our current economy is more and more automated, due to the widespread adoption of software infrastructure. A decision made by software is non-the-less a decision, however the considerations going into framing the decision must match the nature of decision-making delegated to partially or fully autonomous agents.

When relegating parts of the process to software, the initial design phases become more critical as the primary place where concepts of decision analysis can make a difference. Automation necessarily raises the "meta-decision" of whether and to what degree operational decisions should be automated.

Creating a Shared Understanding of the Problem

A key value of the methods used in Decision Analysis is the transparency in the approach, starting from how the problem is framed and continuing throughout the process in how recommendations are explained.

Who participates. Different roles in the analysis

Decision maker

Decision facilitator / analyst

Subject experts

Implementors, Execution

These roles are fluid, and one person, possibly with the assistance of AI "agents" may participate in more than one.

The organizational or personal situation

Solving the right problem. Often the scope of the problem must be re-examined. A narrow scope limits the efficacy of the analysis, and at worst has a pre-conceived conclusion baked into the formulation. What is the point of analysis if it only serves to enforce one's presumption about what should be done?

Developing creative alternatives

Homework, due __

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